

Buyer Agent Master Audit & Verification Report

Complete Stakeholder-Aware AgriNegotiator SRS Architecture & Negotiation Certification

Branch: feature/buyer-agent-verification | **Target:** main | **Repository:** Ritikmehta080905/FarmGenAI | **Status:** 100% Certified (94/94 Tests Passed)

1. Executive Summary & Architectural Scope

This Master Audit unifies the **83 safety/negotiation hardening tests** with the **complete stakeholder-aware AgriNegotiator SRS architecture**. The system enforces strict isolation between **SINGLE_AGENT** (locking permitted_agents=["BUYER"] and strictly preventing accidental downstream execution) and **FULL_SUPPLY_CHAIN** (conditionally invoking Transport, Warehouse, or Processor ONLY when actually required by constraints).

2. Architectural Verification Matrix (10 Core Domains)

Domain	SRS Architectural Invariant	Implementation & Evidence	Status
1. Single vs. Full	SINGLE_AGENT must never invoke downstream; FULL conditionally invokes them.	workflow_policy_gatekeeper in buyer_graph.py locks permitted_agents.	PASS
2. Permitted vs. Required	permitted = allowed services; required = actually needed services.	Explicit contract blocks unpermitted services even if requested by user.	PASS
3. Market Intelligence	Live Mandi + ML trends + transport estimates + RAG context.	buyer_market_context_service.py coordinates Mandi feed + XGBoost + RAG.	PASS
4. Landed Cost Matching	Landed = Base + Freight + APMC Cess. Lowest purchase price ≠ lowest cost.	Local 49 lot wins over remote 48 lot due to highway freight calculation.	PASS
5. RAG vs. Facts	RAG retrieves domain knowledge; live prices come strictly from Mandi APIs.	RAG ChromaDB supplies quality norms; live Agmarknet API supplies modal price.	PASS
6. LangGraph StateGraph	StateGraph multi-node engine decides agent eligibility dynamically.	11-node compiled LangGraph StateGraph with conditional branching.	PASS
7. Event Telemetry	Granular WebSocket / Redis event streaming across procurement stages.	Emits MATCH_FOUND, NEGOTIATION_STARTED, OFFER_RECEIVED, DEAL_SELECTED, etc.	PASS
8. Copilot Guardrails	Human / Copilot overrides cannot bypass P_max, budget, or agent scope.	validate_copilot_buyer_override() rejects P_max breach, budget overrun, agent bypass.	PASS
9. Downstream Escalation	Conditional downstream dispatch for Transport, Warehouse, Processor.	Conditionally calls assign_transport, assign_storage, or processor_service.	PASS
10. PostgreSQL Consistency	Durable state in PostgreSQL with digital contracts and idempotency.	SHA-256 digital contract hash (0x...), TXN-MH-2026-..., 64-char idempotency key.	PASS

3. Test Certification Scorecard (94 / 94 Passed — 100% Pass Rate)

Test Suite File	Coverage Description	Pass / Total	Status
test_buyer_master_srs_architecture.py	Full/Single Supply Chain, Permitted vs Required, Landed Cost, LangGraph, Copilot	11 / 11	PASS (100%)
test_buyer_adversarial_pmax_budget.py	NaN/Inf/Negatives/Strings, P_max override, budget lock, quantity lifecycle, freshness	14 / 14	PASS (100%)
test_buyer_runtime_e2e.py	Full runtime E2E chain, no-deal boundary, crop guardrail, invalid quantity	4 / 4	PASS (100%)
test_05_buyer_agent_extensive.py	Buyer personas, concession curves, utility scoring, memory, contracts	38 / 38	PASS (100%)
test_07_buyer_guardrail_parallel.py	APMC statutory price caps, parallel auto-selection, floor rejection	5 / 5	PASS (100%)
test_09a_buyer_negotiation_orchestration.py	Top-5 selection, state isolation, landed cost calculation, transcripts	22 / 22	PASS (100%)
TOTAL	Complete SRS Architecture & Safety Certification	94 / 94	CERTIFIED

4. Pull Request & Live Verification Links

- **Active PR #5:** <https://github.com/Ritikmehta080905/FarmGenAI/pull/5>
- **Live Application:** <http://localhost:8080/dashboard/buyer>
- **Closed PR:** PR #4 (closed and superseded by PR #5)